Introduction to Software Engineering

Project Proposal & Plan

The student team is required to complete the **Project Proposal & Plan** documentation for the assigned course project, following the attached template.



Software Engineering Department
Faculty of Information and Technology
University of Science

Table of Contents

Right-click here and choose “Update Field” to build the table of contents.

Project Proposal & Plan

Objectives

This document focus on the following topics:

- ✔ Completing the Project Proposal & Plan document with the following sections:
  - Vision
  - Proposed Solution
  - Project Plan
  - Risk Management
  - Tools & Collaboration Setup
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Project Proposal & Plan document.

1 Member Contribution Assessment

ID	Name	Role(s)	Contribution (%)	Evidence
24127397	Nguyễn Lê Huy	*Team lead*	25%	**1.2.1**, **1.2.2**, **1.2.1**
24127061	Ngô Trần Anh Khoa	*Developer Tester*	25%	**1.1.1**, **1.2.2**
24127050	Tất Gia Huy	*Designer*	25%	**1.1.2**, **1.2.2**, **1.2.2**
24127571	Đoàn Minh Triết	*Business analyst*	25%	**1.1.2 1.2.1 1.2.2**

2 Vision

2.1 Preliminary Problem Statement

Currently, the university library provides study rooms for student use; however, the booking process relies on shared Excel files. This manual approach lacks a mechanism for managing concurrent access, frequently leading to data conflicts, unintentional overwriting of information, or double-bookings—especially when many students attempt to reserve rooms during peak hours. Furthermore, the existing system lacks real-time status updates and automated notifications, making it difficult for students to track room availability and hindering library staff from managing resources efficiently.

To address these issues, the proposed solution is a centralized application called "StudySpace," which includes a mobile version (developed using React Native) for students and a web-based dashboard for administrators. The core mechanism for handling concurrent access is a "booking hold" system: when a user selects a time slot, its status is immediately set to "pending/locked" for a brief period, preventing others from reserving the same slot. The system will integrate with the university's Single Sign-On (SSO) for user authentication and utilize a robust backend (built with FastAPI and MongoDB/Firebase) to ensure real-time data synchronization, thereby eliminating data overwrites and optimizing the entire room-booking process.

2.2 Product Position Statement

Element	Content
For	University students and library administrators
Who	Need a reliable, conflict-free way to find, book, and manage study room reservations.
The (product name)	StudySpace
That	Prevents booking conflicts with real-time status updates, automates reminders, and simplifies access via the University SSO.

Unlike	The current manual process using shared Excel spreadsheets, which is prone to double-booking, data overwriting, and requires high administrative effort.
Our product	Provides a robust concurrency control mechanism that temporarily holds slots during transactions, ensuring seamless, real-time room management on both mobile and web platforms.

2.3 Stakeholders and Users

#	Stakeholder / user	Description	Main interests
1	Project sponsor / Library Management	Representatives of the university library or school management board.	Optimizing campus resources, easy monitoring of system operations, and generating usage reports.
2	Student (End User)	University students who need spaces for self-study or group work.	Fast and easy room search, conflict-free booking, and receiving timely automated reminders.
3	System Admin / Librarian	Staff responsible for overseeing the daily operations of the library study rooms.	Easily managing the room catalog (add, edit, lock for maintenance) and overseeing reservations.

2.4 Scope and Key Features

ID	Feature	Description	Priority (M/S/C/W)
F01	User Authentication	Integration with University Authentication System (SSO) and application-managed role permissions.	Must
F02	Room Search & Details	Mobile room search, applying filters, and viewing room details.	Must
F03	Booking & Conflict Prevention	Booking flow with capacity, time validation, and a concurrent-booking prevention mechanism (temporary hold).	Must
F04	Booking Management	Booking history management, cancellation flow, and immediate slot release for rebooking.	Must
F05	Room Administration	Basic room management (add/edit), handling unavailable periods, and reservation oversight.	Must

F06	Automated Notifications	Email/Notification service for booking confirmations, reminders, and cancellations.	Should
F07	Advanced Filters	Advanced room search by specific equipment (e.g., projector, whiteboard) or specific capacity.	Should
F08	Usage Analytics	Basic dashboard for admins to view room usage frequency and statistics.	Could
F09	QR Check-in	QR code check-in/check-out system at the physical room door to confirm attendance.	Could

Out of scope:

- Payment processing or billing systems (this is a free service for students).
- Library book borrowing, returning, or catalog management (handled by a separate library system).

3 Proposed Solution

3.1 Preliminary Architecture

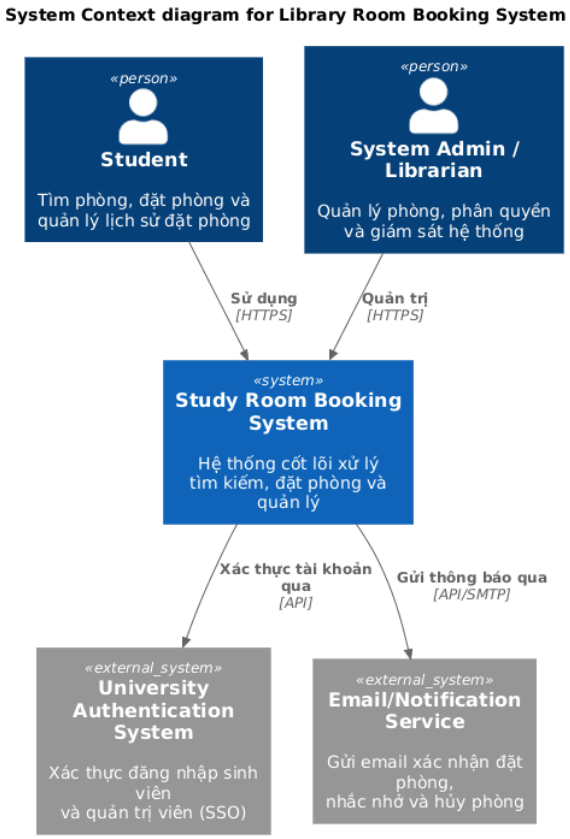
1. Student (User): Interacts with the web application to search for available rooms by date/time/capacity, place bookings, and manage booking history.

2. System Administrator / Librarian: Manages room catalog (add, edit, lock for maintenance), manages user roles, and oversees overall system operations.

3. Study Room Booking System (Core): The central software system that handles user sessions, computes room availability logic, and processes booking transactions.

4. University Authentication System (External): Authenticates students and administrators using their standard university credentials (SSO, Active Directory,...).

5. Email/Notification Service (External): Sends booking confirmations, reminders, and cancellation notices to users (SendGrid, SMTP,...).



3.2 Technology Stack

Layer	Technology	Reason for choice
Front end	TypeScript, React Native	React Native supports cross-platform mobile application development, while TypeScript provides static typing and improves code maintainability and reliability.
Back end	Python, FastAPI	Python is suitable for rapid development, while FastAPI provides a lightweight framework for developing RESTful APIs with type hints and automatic API documentation.
Database	Firebase, MongoDB Atlas	Firebase can provide authentication and cloud-based application services, while MongoDB Atlas provides a managed cloud database for storing application data such as users, rooms, and bookings.
Hosting / deployment	Render, Expo EAS Build	Render is used to deploy the FastAPI backend as a web service, while Expo EAS Build can be used to build and distribute the React Native mobile application for testing and demonstration.
Testing & CI	GitHub Actions	GitHub Actions can automate testing and development workflows when changes are pushed or submitted through pull requests, helping the team detect issues before merging changes.

3.3 Hardware and Operating Environment

Category	*Specification / Environment*	*Purpose*
Development computers	*Laptop/Desktop with at least 8 GB RAM, preferably 16 GB RAM*	*Software development, running IDE, emulator, backend server and development tools*
Mobile devices	*Android smartphone; iOS device may be used for additional testing*	*Testing the mobile application on real devices*
Android emulator	*Android Studio Emulator*	*Testing different screen sizes and Android versions*
Operating System	*Windows 10/11, macOS, or Linux for development*	*Supports the development environment and required tools*

Backend environment	*Python 3.x with FastAPI and Uvicorn*	*Local development and testing of the backend API*
Database environment	*MongoDB Atlas and Firebase cloud services*	*Store and manage application data and authentication services*
Internet connection	*Stable Internet connection*	*Accessing GitHub, cloud services, MongoDB Atlas, Firebase, and deployment platforms*

4 Project Plan

4.1 Process and Phase Gates

The project follows the waterfall model. Each phase ends with a phase gate (a PA submission). After a PA is submitted it becomes a baseline; later changes must be updated in the document and listed in the next PA.

PA	Phase	Due	Exit criteria (fill in)
PA1	Initiation & planning	End of week 3	Reach an agreement on what to do, who will do it, how long it will take, what the risks are, and which tools to use.
PA2	Requirements	End of week 5	Clarify the requirements the system must meet and how users will interact with it.
PA3	Design	End of week 7	Define the organization of the system, data, interfaces, and processing
PA4	Test planning	End of week 8	Prepare testing methods and test cases for mandatory requirements
PA5	Implementation, testing & deployment	End of week 10	Finalize the product, run tests, log bugs, and deliver a demo version
Final	Presentation & demo	Week 11	Presentations, demos, and Q&A

4.2 Work Breakdown Structure (WBS)

[Break the project into phases, deliverables and work packages (at least 3 levels). Each work package must be small enough to be assigned to one owner, estimated and tracked on the task board.]

WBS ID	Task Title	Expected Output
1	**Initiation and Planning**	**Approved project scope, approach, estimates, schedule, roles, risks, and PA1 package.**

1.1	**Problem and Solution Definition**	**Agreed problem statement, target users, project scope, prioritized features, and preliminary solution.**
1.1.1	Define the problem, users, scope, and features	Agreed problem statement, target users, project scope, and prioritized feature list.
1.1.2	Propose the solution and evaluate technologies	Preliminary solution architecture, technology comparison, and context diagram.
1.2	**Project Planning and Setup**	**Reviewed project plan, assigned roles, risk register, and collaboration setup.**
1.2.1	Prepare the WBS, estimates, schedule, and costs	Reviewed WBS, effort estimates, Gantt schedule, and cost estimate.
1.2.2	Assign roles, assess risks, set up tools, and complete PA1	Role assignments, risk register, collaboration setup, and submitted PA1 package.
2	**Requirements**	**Reviewed requirements baseline, use cases, prototype, and PA2 package.**
2.1	**Requirements Baseline**	**Prioritized functional requirements, business rules, quality requirements, and acceptance criteria.**
2.1.1	Specify functional requirements	Prioritized functional requirements for accounts, room search, booking, cancellation, and administration.
2.1.2	Define business rules, quality requirements, and acceptance criteria	Documented booking rules, measurable non-functional requirements, and acceptance criteria.
2.2	**Use Cases and Prototype**	**Use case model, workflow specifications, reviewed low-fidelity prototype, and PA2 package.**

2.2.1	Create the use case model and specifications	Use case diagram and specifications for all Must-have workflows.
2.2.2	Create mockups, review requirements, and complete PA2	Reviewed low-fidelity prototype, traceability review, and PA2 package.
3	**System Design**	**Reviewed architecture, data and interface designs, and PA3 package.**
3.1	**Architecture and Data Design**	**Approved architecture, deployment and component designs, ADRs, and database design.**
3.1.1	Design the architecture, deployment, and key decisions	C4 architecture and deployment views, component design, and ADRs.
3.1.2	Design the database and data constraints	Database schema with integrity constraints and booking-conflict prevention approach.
3.2	**Detailed System Design**	**Class and interaction models, screen specifications, traceability review, and PA3 package.**
3.2.1	Design classes and their interactions	Class model, class descriptions, sequence diagrams, and relevant state diagrams.
3.2.2	Specify screens, check traceability, and complete PA3	Key screen specifications, accessibility checks, requirements traceability, and PA3 package.
4	**Test Planning**	**Reviewed test plan, test cases, coverage review, and PA4 package.**
4.1	**Test Preparation**	**Test plan, prioritized test cases, sample data, requirements coverage, and PA4 package.**
4.1.1	Prepare the test plan	Test scope, environment, entry and exit criteria, responsibilities, and defect process.

4.1.2	Design test cases and prepare test data	Prioritized test cases and sample data using suitable test-design techniques.
4.1.3	Check requirements coverage and complete PA4	Coverage evidence for all Must-have requirements and PA4 package.
5	**Implementation**	**Working mobile application and backend covering the agreed Must-have features.**
5.1	**Platform and Account Foundation**	**Buildable mobile and backend project, database setup, CI workflow, account access, and role permissions.**
5.1.1	Set up the mobile app, back end, database, and CI	Buildable project, database setup, and CI workflow for builds and unit tests.
5.1.2	Implement account access and role permissions	Application-managed account access and role-based permissions.
5.2	**Room Search and Booking**	**Room search, booking validation, booking history, cancellation, and immediate slot release.**
5.2.1	Implement room search, filters, and details	Mobile room search and details connected to the backend.
5.2.2	Implement booking validation and conflict prevention	Booking flow with capacity, time, and concurrent-booking validation.
5.2.3	Implement booking history, cancellation, and slot release	Booking history and cancellation flow that releases a cancelled slot for immediate rebooking.
5.3	**Room Administration**	**Basic room management, unavailable periods, and reservation oversight.**

5.3.1	Implement room administration, maintenance, and reservation oversight	Basic room management, unavailable periods, and reservation oversight.
6	**Testing and Delivery**	**Tested prototype, test evidence, deployment materials, and PA5 package.**
6.1	**Verification and Defect Resolution**	**Executed test results, automated mobile system tests, defect records, regression results, and coverage report.**
6.1.1	Execute tests and record results, defects, and coverage	Test results, defect records, and coverage report.
6.1.2	Implement automated mobile system tests	Automated system tests for selected key use cases and scenarios.
6.1.3	Fix defects and perform regression testing	Resolved defects with regression-test results.
6.2	**Release Package**	**Test deployment, user guide, demo video, retrospective, and PA5 package.**
6.2.1	Deploy the prototype and prepare the guide, video, retrospective, and PA5	Test deployment, user guide, demo video, retrospective, and PA5 package.
7	**Final Presentation**	**Presentation materials and a rehearsed live demonstration.**
7.1	**Presentation Deliverable**	**Final slides, working demo, and rehearsal evidence.**
7.1.1	Prepare slides, the demo, and presentation rehearsals	Final slides, working demo, and rehearsal evidence.

4.3 Estimation

WBS ID	Work package	Proposed owner	Effort (person-hours)	Method / assumptions
1.1.1	Define the problem, users, scope, and features	Business analyst	3	Expert judgement; reuse PA0 and clarify the Excel workflow and mobile scope.
1.1.2	Propose the solution and evaluate technologies	Designer	4	Expert judgement; compare a limited set of options and create a system context diagram.
1.2.1	Prepare the WBS, estimates, schedule, and costs	Team lead	5	Expert judgement; collect member estimates and prepare the WBS and Gantt chart.
1.2.2	Assign roles, assess risks, set up tools, and complete PA1	Team lead	7	Expert judgement; Discord already exists; basic repository and board setup; at least five risks.
2.1.1	Specify functional requirements	Business analyst	5	Expert judgement; cover accounts, room search, booking, cancellation, and administration.
2.1.2	Define business rules, quality requirements, and acceptance criteria	Business analyst	4	Expert judgement; define time, capacity, and cancellation rules and measurable quality requirements.
2.2.1	Create the use case model and specifications	Business analyst	4	Expert judgement; specify all Must-have use cases, with at least six specifications.
2.2.2	Create mockups, review requirements, and complete PA2	Designer	5	Expert judgement; prepare a simple interactive prototype covering the main screens.
3.1.1	Design the architecture, deployment, and key decisions	Designer	5	Expert judgement; C4, component, and deployment diagrams, plus at least three ADRs.
3.2.1	Design classes and their interactions	Designer	6	Expert judgement; six to eight key classes, three to five sequence diagrams, and relevant state diagrams.
3.1.2	Design the database and data constraints	Developer	4	Expert judgement; one database, integrity constraints, and a booking-conflict prevention strategy.
3.2.2	Specify screens, check traceability, and complete PA3	Designer	8	Expert judgement; five to six key screen specifications, accessibility checks, and requirements traceability.
4.1.1	Prepare the test plan	Tester	8	Expert judgement; define scope, environment, entry/exit criteria, responsibilities, and defect handling.
4.1.2	Design test cases and prepare test data	Tester	10	Expert judgement; apply equivalence partitioning, boundary value analysis, and decision tables; specify 15–20 key test cases.
4.1.3	Check requirements coverage and complete PA4	Business analyst	8	Expert judgement; every Must-have requirement must have at least one test case.

5.1.1	Set up the mobile app, back end, database, and CI	Developer	3	Expert judgement; one mobile platform; CI builds the project and runs unit tests.
5.1.2	Implement account access and role permissions	Developer	6	Expert judgement; application-managed accounts without university login integration.
5.2.1	Implement room search, filters, and details	Designer	8	Expert judgement; include mobile screens, API integration, and logic tests.
5.2.2	Implement booking validation and conflict prevention	Team lead	4	Expert judgement; check capacity, time restrictions, and concurrent booking requests.
5.2.3	Implement booking history, cancellation, and slot release	Business analyst	5	Expert judgement; users manage their own bookings and cancelled slots become available immediately.
5.3.1	Implement room administration, maintenance, and reservation oversight	Developer	12	Expert judgement; basic administration without advanced usage analytics.
6.1.1	Execute tests and record results, defects, and coverage	Tester	10	Expert judgement; execute planned test levels and report results and unit-test coverage.
6.1.2	Implement automated mobile system tests	Team lead	10	Expert judgement; automate at least two use cases with two scenarios each.
6.1.3	Fix defects and perform regression testing	Developer	8	Expert judgement; cover normal expected rework; major unforeseen issues use the contingency reserve.
6.2.1	Deploy the prototype and prepare the guide, video, retrospective, and PA5	Team lead	6	Expert judgement; one installable test build, an accessible back end, and a video of no more than ten minutes.
7.1.1	Prepare slides, the demo, and presentation rehearsals	Team lead	8	Expert judgement; total team effort, with every member speaking for at least two minutes.

4.4 Schedule

StudySpace Project Gantt Schedule link: https://vnshort.com/czSP

4.5 Roles and Responsibilities

Role	Main	Backup	Responsibilities
Team lead / Project manager	Nguyễn Lê Huy	Ngô Trần Anh Khoa	Plan, assign tasks, weekly status report, contact point
Business analyst	Đoàn Minh Triết	Tất Gia Huy	Elicit and document requirements, organise reviews
Designer / Architect	Tất Gia Huy	Đoàn Minh Triết	Architecture, class, data and UI design
Developers	Ngô Trần Anh Khoa	Nguyễn Lê Huy	Implement, unit test, review pull requests

		Tất Gia Huy Đoàn Minh Triết	
Tester	Ngô Trần Anh Khoa	Đoàn Minh Triết	Test plan, test cases, system testing, test report

4.6 Human Resources and Cost Estimate

Member	Main Responsibilities	Estimated Effort
Nguyễn Lê Huy	Project planning, WBS, estimation, schedule management, phase gates, and coordination of team activities	50 person-hours
Đoàn Minh Triết	Preliminary problem statement, stakeholder and user analysis, product position statement, scope and key features, and AI usage log	50 person-hours
Tất Gia Huy	Preliminary C4 architecture, risk management, document formatting, and PDF preparation	50 person-hours
Ngô Trần Anh Khoa	Technology stack, hardware and operating environment, tools and collaboration setup, and human resources & cost estimation	50 person-hours

5 Risk Management

ID	Risk	Type	Probability	Loss	RE	Mitigation / contingency	Owner
R1	Key member unavailable during the implementation phase (PA5)	People	3	4	12	Cross-train team members; assign backup roles for critical tasks.	Team lead
R2	Unclear library business rules (max hours, penalty for no-shows)	Requirements	3	3	9	Clarify rules with stakeholders early; document clearly in SRS.	Business analyst
R3	High concurrency during exam weeks causing double-booking issues	Technology	4	4	16	Implement strict database transaction locking and conduct stress testing.	Developer
R4	Cross-browser compatibility and responsive UI issues on the web application	Technology / UI	3	4	12	Design responsive web mockups and test thoroughly on major browsers	Designer
R5	Unauthorized access to admin features to lock rooms	Security	2	5	10	Conduct robust role-based access control (RBAC) security testing.	Tester

6 Tools & Collaboration Setup

The team uses the following tools and collaboration practices to support project management, communication, version control, and documentation.

Tool / Platform	Purpose	Team Usage
Moodle	Course communication and submission	Receive course announcements and submit PA documents
Discord	Team communication	Discuss project tasks, coordinate work, and share updates
GitHub Projects	Task management	Create and track WBS work packages, assign owners, set due dates, and monitor progress
GitHub Repository	Version control and source code management	Store source code, documentation, and project deliverables
GitHub Actions	Testing and CI	Automate testing and development workflows when changes are pushed or submitted through pull requests

- ✔ Weekly status meeting (15 minutes) and weekly status report (template provided), committed to /docs/management/weekly every week from week 2 to week 10. The team chooses its weekly deadline and states it here.

Repository structure:

Folder	Content

/src	Source code
/docs/management	Project plan; weekly status reports in /docs/management/weekly
/docs/requirements	Vision, SRS, use cases, mockups
/docs/design	Architecture (C4), ADRs, UML models, UI design
/docs/test	Test plan, test cases, test reports
/docs/ai-usage	AI usage logs
/pa/PA1 … /pa/PA5	One sub-folder per PA submission (PDF + source files)

7 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Lê Huy	ChatGPT	Understanding PA1 requirements. Preparing the WBS, estimates, Gantt, and task assignments. Drafting the weekly status report and learning Git commands .	Checked against the course guidelines and PA1 template. Matched with the templates, schedule, and team roles; adjusted tasks and owners. commandsChecked against team updates and terminal output; corrected report details and file path.
2	Ngô Trần Anh Khoa	ChatGPT	Drafted Technology Stack, Hardware & Operating Environment, Human Resources, and Tools & Collaboration Setup.	Reviewed against PA0/PA1 requirements and adjusted to fit the project.
3	Đoàn Minh Triết	Gemini, Claude	Translate and format my defined Vietnamese concepts (Problem statement, "Booking Hold" solution, MoSCoW features) into formal English business text.	I reviewed the English output to ensure the AI strictly followed my technical logic and did not add any features outside my WBS. I adjusted the wording to match the team's exact constraints.
4	Tất Gia Huy	ChatGPT	Drafted the Preliminary Architecture (C4 System Context) description and generated Mermaid/PlantUML code for the diagram. Drafted the	How the output was verified and changed: Reviewed against the project context (Library Room Booking System), adjusted risk owner roles to match the actual team structure.

			Risk Management table with project risks mapped to team roles.	